

Banking Risk Analytics System – SQL Report

1. Introduction

This project simulates a real-world banking system using PostgreSQL. The objective is to design a structured relational database and perform analytical operations to extract meaningful business insights.

The system focuses on understanding customer behavior, transaction trends, loan risks, and branch performance.

2. Objectives

- Design a normalized database schema for banking operations
- Establish relationships between customers, accounts, transactions, and loans
- Perform business-driven analysis using SQL
- Identify high-value customers and potential risk areas

3. Database Design

The database consists of the following core entities:

- **Customers** → Stores personal and financial details
- **Accounts** → Represents bank accounts linked to customers
- **Transactions** → Tracks all financial activities
- **Loans** → Contains loan-related information
- **Loan Payments** → Tracks repayment behavior
- **Branches** → Represents bank locations
- **Employees** → Maps employees to branches

The schema follows a **relational model with one-to-many relationships**, ensuring data consistency and integrity.

4. Relationships Overview

- One customer can have multiple accounts
- One account can have multiple transactions
- One customer can have multiple loans
- One loan can have multiple payments
- One branch can manage multiple accounts and employees

5. Analytical Insights

Customer Analysis

- Identified high-value customers based on total account balance
- Segmented customers based on financial strength

Transaction Analysis

- Analyzed monthly transaction trends to track activity growth

- Compared debit and credit patterns to understand spending behavior
- Identified top spending customers

Loan & Risk Analysis

- Detected customers with defaulted loans
- Measured total loan exposure per customer
- Identified missed or delayed payments as risk indicators

Branch Performance

- Evaluated number of accounts handled by each branch
- Compared total balances across branches to identify high-performing locations

6. Advanced Analytical Techniques

- Ranking customers based on financial value
- Identifying high-value customers using structured query logic
- Applying multi-level analysis using relational joins

7. Key Learnings

- Strong understanding of relational database design
- Practical exposure to PostgreSQL
- Ability to translate business problems into analytical queries
- Hands-on experience with financial data modeling

8. Conclusion

This project demonstrates the ability to build a structured banking database and derive meaningful insights that can support decision-making in financial institutions.

It highlights both technical SQL skills and business-oriented analytical thinking, making it relevant for Data Analyst and Business Analyst roles.

9. Future Enhancements

- Integration with Python for advanced analytics
- Dashboard development using Power BI
- Expansion of dataset for real-world scalability